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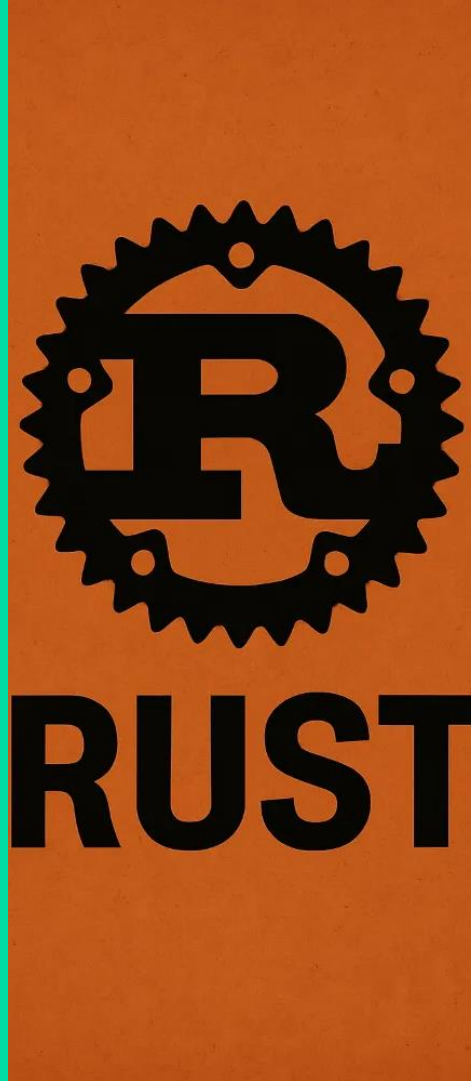
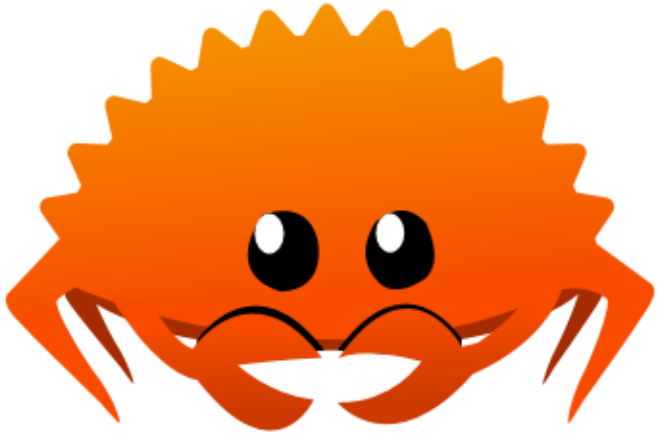
ECM2433: The C Family

Lecture 01: Tooling and Getting Started



Why Rust?

- Memory bugs: use-after-free, double-free, buffer overflows
- C++ added RAII and smart pointers: optional, not enforced
- Rust: compile-time ownership + runtime bounds checks
- C/C++ performance, no garbage collector



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The Rust Book

- " The Rust Programming Language" : primary reference
- Free online: https://doc.rust-lang.org/stable/book/
- Chapter citations throughout: [Book ch X.X]
- Each video ends with required reading

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Installation and Verification

- Already installed on university machines
- Verify: `rustc --version`
- Optional: install on your laptop [Book ch 1.1]
- `rustup doc`: opens local docs offline
- `rustup doc --book:` opens the Rust Book offline

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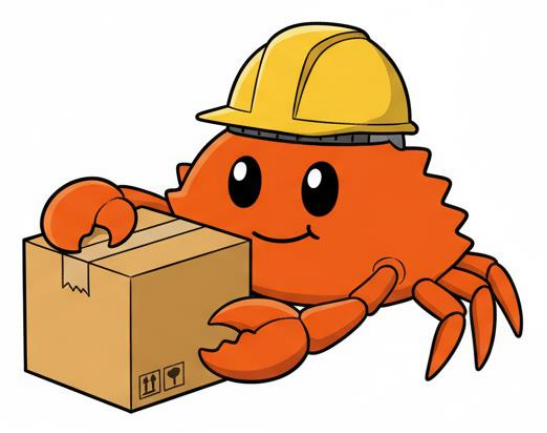
What is Cargo?

- Build system + package manager

- Replaces Makefiles, CMake, manual dependencies

- `cargo new`: create project
- `cargo build`: compile
- `cargo run`: compile and run



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Creating a Project

- `cargo new hello_rust`

- Creates:
- `src/main.rs`: your code
- `Cargo.toml`: project config
- `.git/`: initialised if Git is installed

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Project Files

- `main.rs`: entry point, starts with hello world
- `Cargo.toml`: name, version, edition, dependencies
- `Cargo.lock`: exact dependency versions (auto-generated)

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Building and Running

- `cargo build`: compile to target/debug/
- `cargo check`: check without binary (faster)
- `cargo run`: compile and run

Code Quality Tools

- `cargo fmt`
  - formats to standard Rust style
- `cargo clippy`
  - catches mistakes, suggests improvements
- Run both before submitting work

main.rs U

```rust
fn is_even(n:i32)->bool{
if n%2==0{ true }else{ false }
}

▶ Run | ⚙ Debug
fn main(){
println!("{}",is_even(4));
}
```

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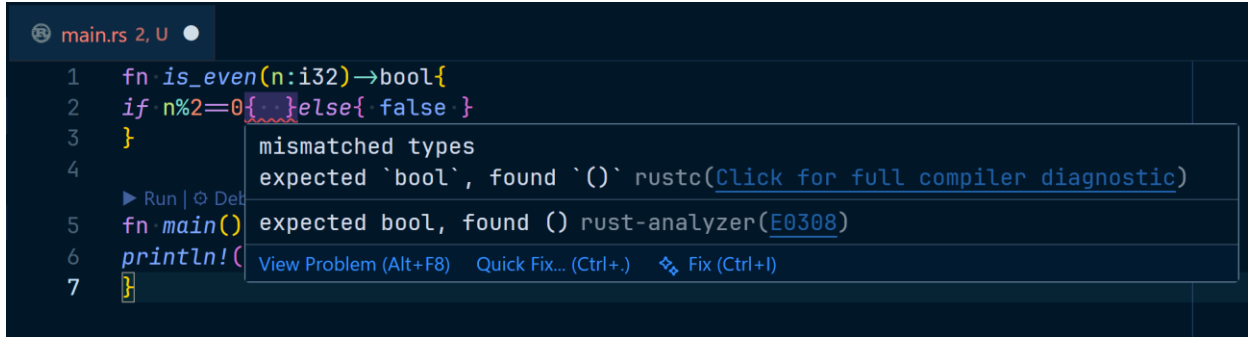
IDE Support: rust-analyzer

- `rust-analyzer`: the Rust language server
- Autocomplete, inline errors, go-to-definition, type hints
- VS Code: install "rust-analyzer" extension
- Compiler errors as you type
- [Book Appendix D]



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Recap

- Rust: memory safety (compile-time + runtime), C/C++ performance
- Cargo: `new`, `build`, `check`, `run`
- Project structure: `src/main.rs`, `Cargo.toml`
- Tools: `rust-analyzer`, `cargo fmt`, `cargo clippy`
- Required reading: Book ch 1.1, 1.2, 1.3, Appendix D

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